

Quantum Operations, Quantum Noise

Vivek Soorya Maadoori

Contents

1. Quantum Operations	4
-----------------------	---

1. Quantum Operations

Exercise 8.1: (Unitary evolution as a quantum operation) Pure states evolve under unitary transforms as $|\psi\rangle \rightarrow U|\psi\rangle$. Show that, equivalently, we may write $\rho \rightarrow \varepsilon(\rho) \equiv U\rho U^\dagger$, for $\rho = |\psi\rangle\langle\psi|$.

Solution:

(1) If pure states evolve as $|\psi\rangle \rightarrow U|\psi\rangle$, then their duals evolve like so $\langle\psi| \rightarrow \langle\psi|U^\dagger$.

(2) Now consider how ρ would evolve:

$$\rho = |\psi\rangle\langle\psi| \rightarrow U|\psi\rangle\langle\psi|U^\dagger = U\rho U^\dagger.$$

■

Exercise 8.2: (Measurement as a quantum operation)

Recall from Section 2.2.3 (on page 84) that a quantum measurement with outcomes labeled by m is described by a set of measurement operators M_m such that

$$\sum_m M_m^\dagger M_m = I.$$

Let the state of the system immediately before the measurement be ρ . Show that for

$$\varepsilon_m(\rho) \equiv M_m \rho M_m^\dagger,$$

the state of the system immediately after the measurement is

$$\frac{\varepsilon_m(\rho)}{\text{tr}(\varepsilon_m(\rho))}.$$

Also show that the probability of obtaining this measurement result is $p(m) = \text{tr}(\varepsilon_m(\rho))$.

Solution: First let's show the post measurement state of the density operator,

(1) Note that $\varepsilon_m(\rho) \equiv M_m \rho M_m^\dagger = M_m |\psi\rangle\langle\psi| M_m^\dagger$

(2) Post measurement state is $\frac{M_m |\psi\rangle}{\sqrt{\langle\psi| M_m^\dagger M_m |\psi\rangle}}$

(3) Post measurement state of the dual is $\frac{\langle\psi| M_m^\dagger}{\sqrt{\langle\psi| M_m^\dagger M_m |\psi\rangle}}$

(4) Thus, the post measurement state of ρ is $\frac{M_m |\psi\rangle\langle\psi| M_m^\dagger}{\langle\psi| M_m^\dagger M_m |\psi\rangle}$ ¹

(5) Note that the numerator is $\varepsilon_m(\rho)$ and that the denominator is $\text{tr}(\varepsilon_m(\rho))$.

Now, let's see the probability

(1) Note that the probability of measuring m after measurement is given by $p(m) = \langle\psi| M_m^\dagger M_m |\psi\rangle$

¹Assuming the measurement operator commutes with its adjoint. Does it?

(2) Note that this is $\text{tr}(\varepsilon(\rho))$

■